

Part 1

```
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls
logs pipeline.conf processed_data source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$
```

Part 2

```
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ chmod700 source_data
chmod700: command not found
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ chmod 700 source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls -a
. .. logs pipeline.conf processed_data source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls -la
total 20
drwxr-xr-x 5 subzero subzero 4096 Feb 11 09:49 .
drwxr-xr-x 4 subzero subzero 4096 Feb 11 09:47 ..
drwxr-xr-x 2 subzero subzero 4096 Feb 11 09:48 logs
-rw-r--r-- 1 subzero subzero  0 Feb 11 09:49 pipeline.conf
drwxr-xr-x 2 subzero subzero 4096 Feb 11 09:48 processed_data
drwx----- 2 subzero subzero 4096 Feb 11 09:48 source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$
```

Part 3

```
subzero@PC: ~/linux-data-en X + v
subzero@PC:~/linux-data-engineering-lab$ cd ./
subzero@PC:~/linux-data-engineering-lab$ ls
README.md  datacorp_pipeline
subzero@PC:~/linux-data-engineering-lab$ cd datacorp_pipeline
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls
logs  pipeline.conf  processed_data  source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ export ETL_STAGE="staging"
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ echo $ETL_STAGE
staging
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$
```

Part 4

```
subzero@PC: ~/linux-data-en X + v
subzero@PC:~/linux-data-engineering-lab$ cd ./
subzero@PC:~/linux-data-engineering-lab$ ls
README.md  datacorp_pipeline
subzero@PC:~/linux-data-engineering-lab$ cd datacorp_pipeline
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls
logs  pipeline.conf  processed_data  source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ export ETL_STAGE="staging"
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ echo $ETL_STAGE
staging
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ sleep 500 &
[1] 723
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ pgrep sleep
723
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ kill 723
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ pgrep sleep
[1]+  Terminated                  sleep 500
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$
```

Part 5

```
W: Some index files failed to download. They have been ignored, or old ones used instead.
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ sudo apt install htop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnl-3-200 libnl-genl-3-200
Suggested packages:
  lm-sensors strace
The following NEW packages will be installed:
  htop libnl-3-200 libnl-genl-3-200
0 upgraded, 3 newly installed, 0 to remove and 204 not upgraded.
Need to get 238 kB of archives.
After this operation, 665 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-genl-3-200 amd64 3.7.0-0.3build1.1 [12.2 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]
Fetched 238 kB in 2s (99.8 kB/s)
Selecting previously unselected package libnl-3-200:amd64.
(Reading database ... 40797 files and directories currently installed.)
Preparing to unpack .../libnl-3-200-3.7.0-0.3build1.1_amd64.deb ...
Unpacking libnl-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package libnl-genl-3-200:amd64.
Preparing to unpack .../libnl-genl-3-200-3.7.0-0.3build1.1_amd64.deb ...
Unpacking libnl-genl-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package htop.
Preparing to unpack .../htop-3.3.0-4build1_amd64.deb ...
Unpacking htop (3.3.0-4build1) ...
Setting up libnl-3-200:amd64 (3.7.0-0.3build1.1) ...
Setting up libnl-genl-3-200:amd64 (3.7.0-0.3build1.1) ...
Setting up htop (3.3.0-4build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
```

```
subzero@PC: ~/linux-data-en x + v - o x

0[ 0.0%] 4[ 0.0%]
1[ 0.0%] 5[ 0.7%]
2[ 0.0%] 6[ 0.0%]
3[ 0.0%] 7[ 0.0%]
Mem[|||||] 378M/7.66G Tasks: 25, 20 thr, 0 kthr; 1 running
Swp[ 0K/2.00G] Load average: 0.14 0.11 0.03
Uptime: 02:02:48

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1145 subzero 20 0 5652 4224 3328 R 0.7 0.1 0:00.22 htop
1 root 20 0 21856 12268 9196 S 0.0 0.2 0:02.60 /sbin/init
2 root 20 0 3120 1920 1920 S 0.0 0.0 0:00.48 /init
6 root 20 0 3136 1920 1920 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
7 root 20 0 3136 1920 1920 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
8 root 20 0 3120 1920 1920 S 0.0 0.0 0:00.00 /init
55 root 19 -1 66844 15540 14516 S 0.0 0.2 0:01.67 /usr/lib/systemd/systemd-journald
103 root 20 0 25136 6144 4864 S 0.0 0.1 0:01.63 /usr/lib/systemd/systemd-udev
164 systemd-re 20 0 21452 12672 10496 S 0.0 0.2 0:00.35 /usr/lib/systemd/systemd-resolved
168 systemd-ti 20 0 91020 7680 6784 S 0.0 0.1 0:00.64 /usr/lib/systemd/systemd-timesyncd
172 systemd-ti 20 0 91020 7680 6784 S 0.0 0.1 0:00.00 /usr/lib/systemd/systemd-timesyncd
177 root 20 0 4236 2560 2432 S 0.0 0.0 0:00.07 /usr/sbin/cron -f -P
178 messagebus 20 0 9656 4864 4352 S 0.0 0.1 0:00.98 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
190 root 20 0 17964 8448 7552 S 0.0 0.1 0:00.43 /usr/lib/systemd/systemd-logind
193 root 20 0 1714M 12672 10240 S 0.0 0.2 0:00.11 /usr/libexec/wsl-pro-service -vv
197 root 20 0 3160 1920 1792 S 0.0 0.0 0:00.01 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
213 root 20 0 3116 1792 1664 S 0.0 0.0 0:00.01 /sbin/agetty -o -p -- \u --noclear - linux
216 syslog 20 0 217M 5376 4352 S 0.0 0.1 0:00.09 /usr/sbin/rsyslogd -n -iNONE
224 root 20 0 104M 22400 13184 S 0.0 0.3 0:00.23 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
226 root 20 0 1714M 12672 10240 S 0.0 0.2 0:00.33 /usr/libexec/wsl-pro-service -vv
227 root 20 0 1714M 12672 10240 S 0.0 0.2 0:00.13 /usr/libexec/wsl-pro-service -vv
228 root 20 0 1714M 12672 10240 S 0.0 0.2 0:00.00 /usr/libexec/wsl-pro-service -vv
229 root 20 0 1714M 12672 10240 S 0.0 0.2 0:00.13 /usr/libexec/wsl-pro-service -vv
F1Help F2Setup F3Search F4Filter F5Free F6SortBy F7Nice F8Nice F9Kill F10Quit
```

Part 6

```
subzero@PC: ~/linux-data-en x + v
none 3.9G 0 3.9G 0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none 3.9G 4.0K 3.9G 1% /mnt/wsl
drivers 238G 187G 52G 79% /usr/lib/wsl/drivers
/dev/sdd 1007G 2.0G 954G 1% /
none 3.9G 80K 3.9G 1% /mnt/wslg
none 3.9G 0 3.9G 0% /usr/lib/wsl/lib
rootfs 3.9G 2.7M 3.9G 1% /init
none 3.9G 516K 3.9G 1% /run
none 3.9G 0 3.9G 0% /run/lock
none 3.9G 0 3.9G 0% /run/shm
none 3.9G 76K 3.9G 1% /mnt/wslg/versions.txt
none 3.9G 76K 3.9G 1% /mnt/wslg/doc
C:\ 238G 187G 52G 79% /mnt/c
tmpfs 785M 20K 785M 1% /run/user/1000
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ df -h > logs/disk_report.txt
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ cd logs
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline/logs$ ls
disk_report.txt
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline/logs$ cat disk_report.txt
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0 3.9G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K 3.9G  1% /mnt/wsl
drivers          238G  187G  52G  79% /usr/lib/wsl/drivers
/dev/sdd        1007G   2.0G 954G  1% /
none            3.9G  80K 3.9G  1% /mnt/wslg
none            3.9G  0 3.9G  0% /usr/lib/wsl/lib
rootfs           3.9G  2.7M 3.9G  1% /init
none            3.9G  516K 3.9G  1% /run
none            3.9G  0 3.9G  0% /run/lock
none            3.9G  0 3.9G  0% /run/shm
none            3.9G  76K 3.9G  1% /mnt/wslg/versions.txt
none            3.9G  76K 3.9G  1% /mnt/wslg/doc
C:\             238G  187G  52G  79% /mnt/c
tmpfs           785M  20K 785M  1% /run/user/1000
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline/logs$ S
```

Part 7

```
subzero@PC: ~/linux-data-en x + v
GNU nano 7.2 daily_maintenance.sh
#!/bin/bash

DATE=$(date +%F)

echo "starting daily maintenance on $DATE ..."

SOURCE_DIR="/home/subzero/linux-data-engineering-lab/datacorp_pipeline/source_data"
ls -l "$SOURCE_DIR"

[ Read 12 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify
^C Location  ^M-U Undo    ^M-A Set Mark
^_ Go To Line ^M-E Redo    ^M-G Copy
```

```
daily_maintenance.sh logs pipeline.conf processed_data source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ nano daily_maintenance.sh
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ chmod 777 daily_maintenance.sh
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ls -l
total 16
-rwxrwxrwx 1 subzero subzero 187 Feb 17 16:01 daily_maintenance.sh
drwxr-xr-x 2 subzero subzero 4096 Feb 11 12:00 logs
-rw-r--r-- 1 subzero subzero 0 Feb 11 09:49 pipeline.conf
drwxr-xr-x 2 subzero subzero 4096 Feb 11 09:48 processed_data
drwx----- 2 subzero subzero 4096 Feb 11 09:48 source_data
subzero@PC:~/linux-data-engineering-lab/datacorp_pipeline$ ./daily_maintenance.sh
starting daily maintenance on 2026-02-17 ...
total 0
```